

Predicting NIRF Engineering relative sub-scores from the submitted-data PDF

A test-only harness: train on 2023–2024, select by leave-one-institute-out cross-validation, evaluate once on untouched 2025.

Summary

Six relative sub-scores were modelled end to end: the model inputs were extracted from the institute's own submitted-data PDF, and the target was read by OCR from the published score graph. Model choice was made on 2023 and 2024 only. On 2025 the median absolute error across the six is 0.67 marks; four of the six come in under a mark, and the two that do not are FPPP (1.17) and FRU (1.44) — both cases where a per-faculty or per-student ratio inherits the volatility of its denominator. The sum of the six sub-scores (125 marks in total) is reproduced to 2.00 marks MAE, R^2 0.974, Spearman 0.973. One result is a clean negative and is reported as such: the prescribed SS inputs are scale-free ratios and cannot reproduce a sub-score that scales with institute size (section 5.4).

Parameter	Max	Inputs	Selected model	CV MAE	2025 n	2025 MAE	2025 RMSE	2025 R^2	bias	≤ 0.5	≤ 1	≤ 2	ρ
ESCS	20	1	Isotonic / pooled	0.83	88.0	0.69	0.90	0.928	-0.12	43%	75%	95%	0.954
GPHD	20	1	PreviousYear / per_year	0.78	86.0	0.65	0.89	0.960	+0.00	56%	80%	97%	0.971
GMS	25	1	PreviousYear / per_year	0.41	86.0	0.54	0.74	0.962	-0.48	57%	90%	98%	0.986
FPPP	10	2	PreviousYear / per_year	0.27	86.0	1.17	1.44	0.692	-1.17	27%	47%	78%	0.976
SS	20	2	PreviousYear / per_year	0.68	86.0	0.57	0.94	0.931	-0.22	69%	78%	93%	0.960
FRU	30	2	AdditiveLog / per_year	0.99	85.0	1.44	1.84	0.874	+1.04	13%	34%	80%	0.958

Errors in marks on the published sub-score scale, 2025. IPR is absent: see section 3.4.

1. Question and protocol

The question is whether a relative sub-score can be predicted from the raw material an institute already publishes — its own NIRF submitted-data PDF — without access to the scoring committee's workings. The protocol is fixed before any 2025 label is read:

Train 2023 and 2024. **Select** leave-one-institute-out (LOIO) cross-validation over those two years only. **Test** 2025, scored once, after the choice was frozen.

No 2025 label entered the feature extractor, the model families, the strategies, the selection rule, or any threshold. The 2025 OCR targets were validated against an independent reference dataset held in the repository (1,224 of 1,224 comparable sub-score cells matched exactly), so the labels are trustworthy even though they were never used for tuning.

2. Data

Year	Ranking rows	PDFs	Score graphs	OCR rows	OCR failures
2023		100	100	100	0
2024		100	100	100	0
2025		90	90	90	0

2025 is cut at the top 90 to match the repository's reference dataset. Graphs are PNG for 2023 and JPEG for 2024–2025. Every downloaded file parsed; the failure log is empty. Feature coverage after extraction, per year:

Parameter	Inputs	2023 usable	2024 usable	2025 usable
ESCS	escs_pct	97	97	88
GPHD	gphd_avg	99	100	90
GMS	gms_median_salary	100	100	90
FPPP	fppp_sponsored_per_faculty, fppp_consultancy_per_faculty	98	99	90
SS	ss_x1, ss_x2	100	100	90
FRU	fru_capital_per_student, fru_operational_per_student	96	96	85

Nothing is imputed. A row missing any one input drops out of that parameter and the reason is logged in harness/out/extract_log.csv.

3. Input features

3.1 Definitions as implemented

Each input is derived only from figures printed in the institute's own PDF. Where the brief and the published graph disagreed, the disagreement is recorded rather than silently resolved — see 3.3.

Parameter	Definition
ESCS	$100 \times \Sigma(\text{UG fee reimbursement from government} + \text{institution} + \text{private bodies}) \div \Sigma(\text{UG rows: total students})$
GPHD	mean of the six printed “Ph.D students graduated” cells (3 financial years $\times$ full-time / part-time)
GMS	graduating-student-weighted mean of the printed median-salary figures over every (programme, year) placement cell
FPPP	$x1 = \text{mean}(3 \text{ y of sponsored "Total Amount Received"}) \div \text{faculty count}$; $x2 = \text{consultancy, same treatment}$
SS	$x1 = \text{NE} \div \text{NT}$, $x2 = \text{Np} \div \text{NT}$, where NT is the sanctioned intake taken per programme in that programme's own most recent year for which the placement count is reported
FRU	$\text{BC} = \text{mean}(3 \text{ y of the four capital-expenditure line items}) \div \text{total student strength}$; BO = operational, same treatment
IPR	not extracted — see 3.4

SS needs a note. The placement section reports first-year admitted counts only for older intake years, and the year it reports differs per programme, so NT is taken per programme in its own most recent reported year. Only programmes that actually report an admitted count contribute to either side of NE/NT, which keeps the ratio internally consistent. 252 of 290 institute-years report the count for three programmes or fewer; that thin coverage is logged per row.

3.2 Parser robustness

The source PDFs are not machine-uniform. Two failure modes dominated and both are handled explicitly rather than by dropping rows:

Split tables. pdfplumber sometimes splits a header into a separate table, leaving the data table headerless. Each financial block therefore falls back to the document-level financial-year column order discovered from whichever financial table kept its header.

Broken row alignment. A number of source PDFs wrap “Total Amount Received” over two lines and print the figures vertically offset, which makes pdfplumber drop the row even though the figures sit in clean columns. Those amounts are read off the word layer instead, anchored to the label and attributed to the right section — carrying the section across page breaks, because the figures routinely spill onto the next page while the heading stays behind. This lifted FPPP sponsored coverage from 62/68/73 to 99/99/90 across 2023/2024/2025, and every recovered value was checked against the rendered text.

3.3 Two places the brief and the published graph disagree

Maxima. The brief quotes SS 15, IPR 10, ESCS 20 and a different GUE split. The published score graphs are the authority for the target scale, so the official maxima are used for the relative conversion (SS 20, IPR 15, FPPP 10, GMS 25, GPHD 20, FRU 30, ESCS 20) and are identical across all three years. Using the brief's SS maximum of 15 would understate every SS error by a third.

Inputs that are ratios but targets that are not. Section 5.4 shows this is not hypothetical: for SS it costs essentially all of the signal.

3.4 IPR: no compliant input exists

IPR is scored from patents granted and published in the year of assessment. The PDFs NIRF publishes as the submitted-data record do not carry a patents block at all — they stop after the student and faculty profile — so there is no way to build the prescribed inputs from the material the model is allowed to see. IPR is therefore excluded from the study rather than approximated, and it is excluded from the combined score as well. Supplying the institute's own IPR return, or NIRF's internal patents data, would make the parameter testable; nothing else about the harness would need to change.

4. Models and how one was chosen

Targets are fitted as relative scores in $[0, 1]$ — the printed sub-score divided by its official maximum — because every prescribed family is monotone in its input. Metrics are reported back on the printed absolute scale, which is the unit the graph is read in.

Class	Families
single input	isotonic · log-linear · capped linear · power law
two inputs	additive log · additive isotonic
baselines	mean of 2023+2024 · previous year's published sub-score
strategies	pooled (one fit over 2023+2024) · per-year (most recent labelled year) · hybrid (50/50 blend of the two)

Every candidate is scored on an identical set of rows. This matters: the previous-year baseline cannot predict a 2023 row, so scoring it over 2023–2024 against models also scored over 2023–2024 would compare different populations. Selection therefore uses only the rows all candidates can predict, and the wider 2023–2024 CV for the fitted families is kept in `harness/out/cv_table_2023_2024.csv`. The two tables agree on every winner.

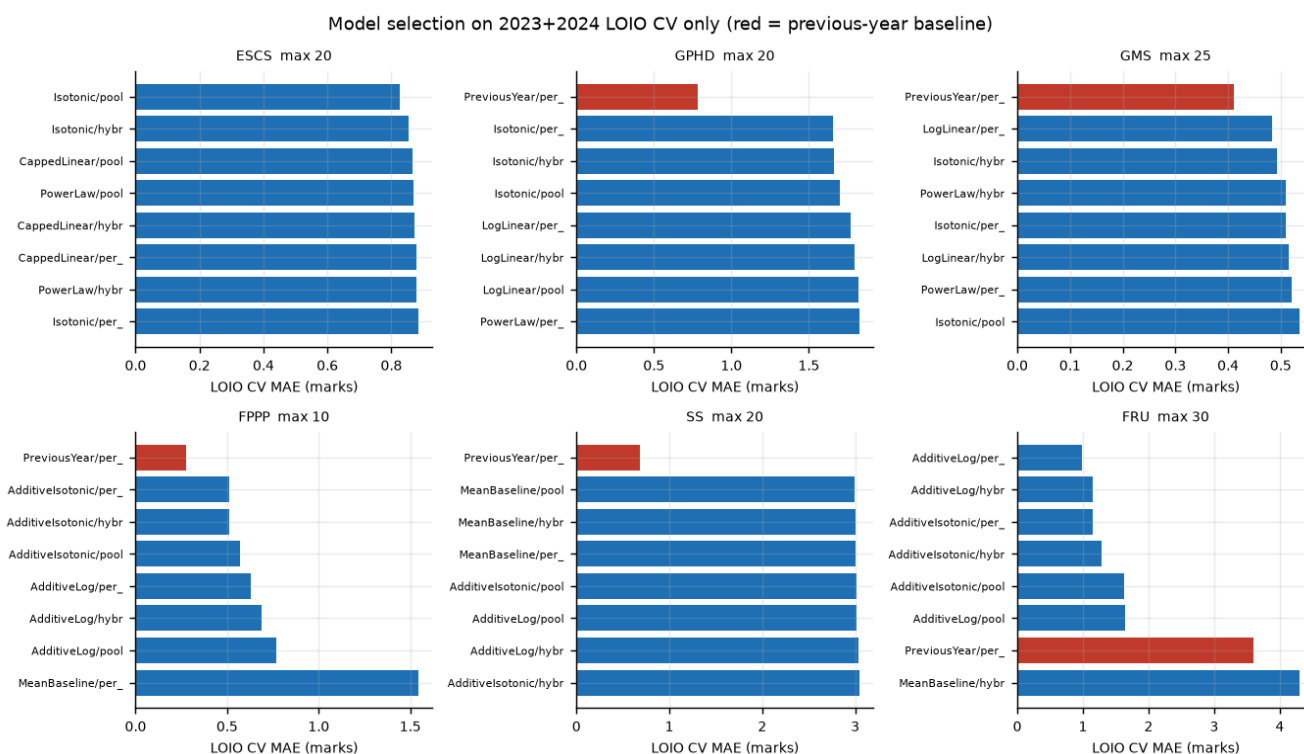


Figure 1 — the eight best candidates per parameter, ranked by 2023–2024 LOIO CV MAE. The previous-year baseline (red) wins outright for four parameters.

5. Results on 2025

5.1 Against the mean baseline

Parameter	Max	Model MAE	Mean-baseline MAE	Improvement	Model RMSE	Model R ²	bias	worst error	ρ
ESCS	20	0.69	2.86	76%	0.90	0.928	-0.12	2.57	0.954
GPHD	20	0.65	3.49	81%	0.89	0.960	+0.00	4.04	0.971
GMS	25	0.54	3.08	82%	0.74	0.962	-0.48	2.93	0.986
FPPP	10	1.17	2.05	43%	1.44	0.692	-1.17	3.42	0.976
SS	20	0.57	3.02	81%	0.94	0.931	-0.22	2.87	0.960
FRU	30	1.44	4.37	67%	1.84	0.874	+1.04	8.48	0.958

Mean baseline = the mean 2023+2024 sub-score for that parameter, applied to every 2025 institute. A negative improvement would mean the model is worse than a constant.

5.2 Accuracy and rank agreement

2025 predicted vs published sub-score, by parameter

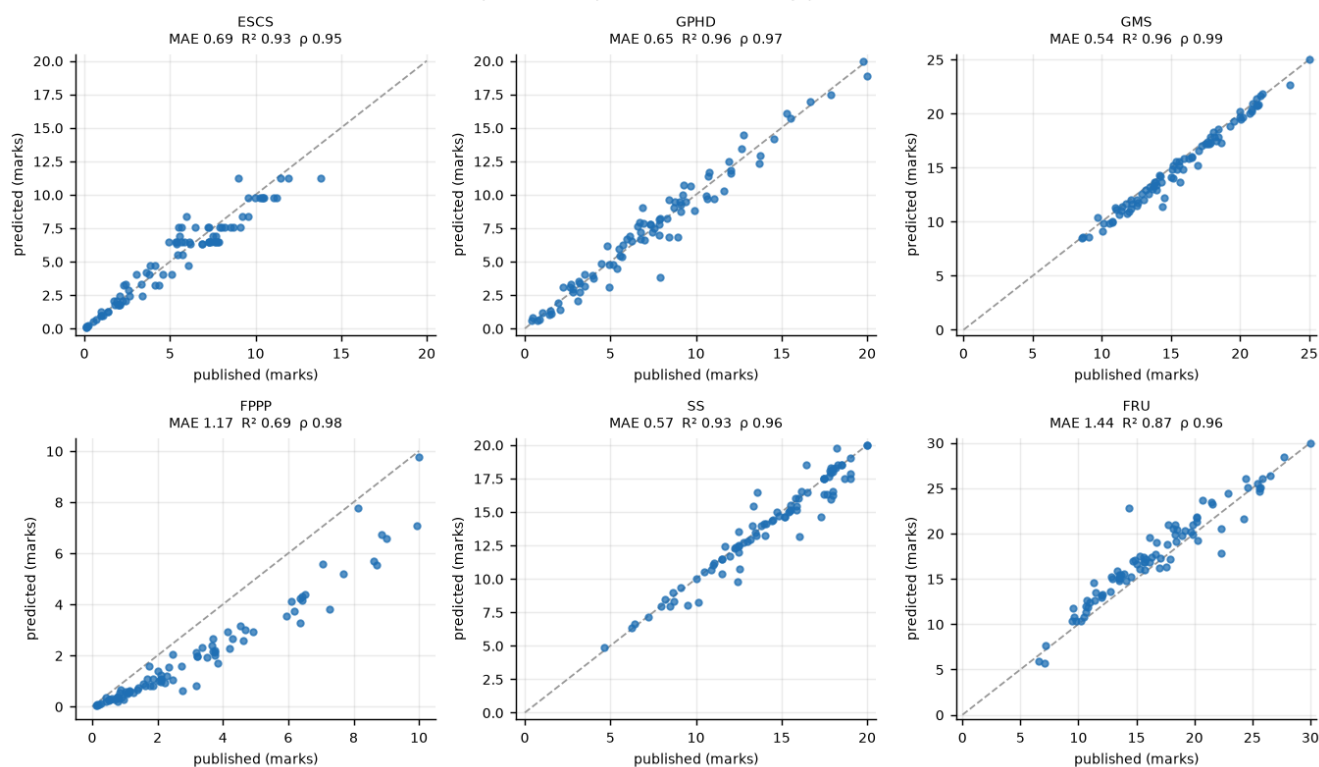


Figure 2 — every scored 2025 institute-year. Dashed line is exact agreement.

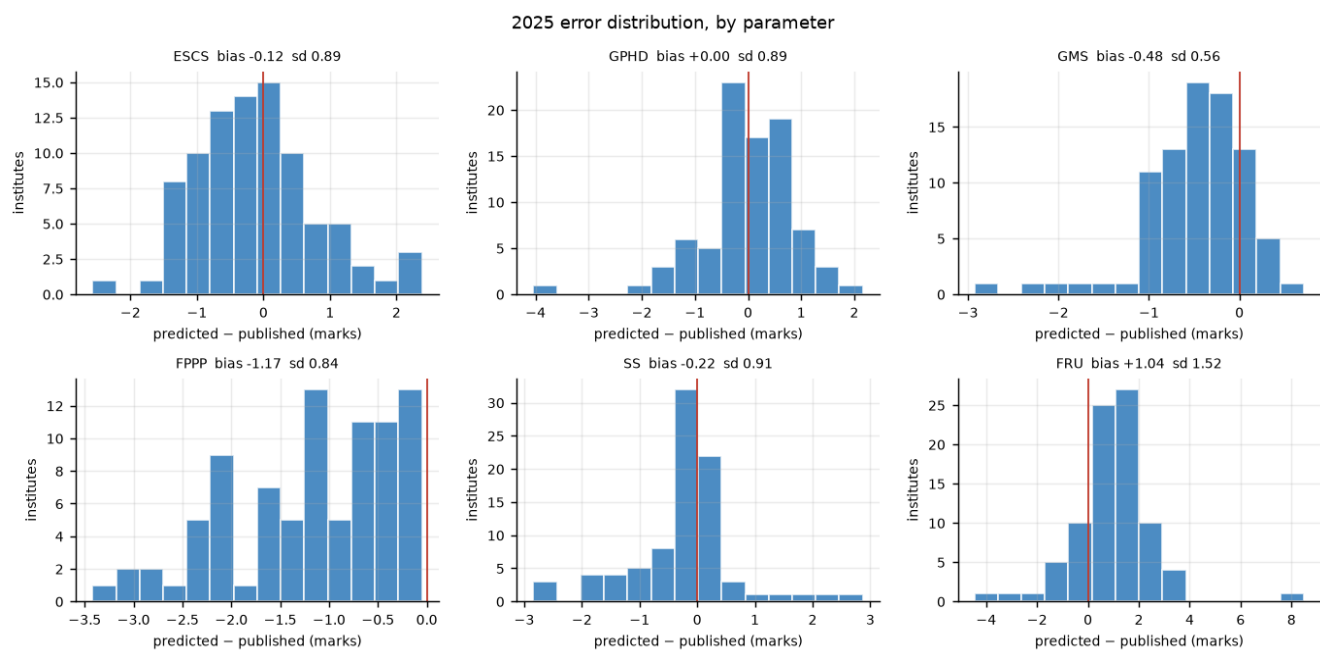


Figure 3 — error distributions. Bias is the signed mean error; a red line marks zero.

5.3 The five worst 2025 errors per parameter

ESCS (max 20)

Institute	Rank	Published	Predicted	error
National Institute of Technology Srinagar	73	13.8	11.23	2.57
Maulana Azad National Institute of Technolog	81	5.97	8.352	2.382
Jamia Millia Islamia	24	9.02	11.23	2.21
COEP Technological University	90	5.51	7.568	2.058
Jadavpur University	18	5.69	7.576	1.886

GPHD (max 20)

Institute	Rank	Published	Predicted	error
Madan Mohan Malaviya University of Technolog	60	7.9	3.86	4.04
Institute of Chemical Technology	41	6.88	9.04	2.16
Delhi Technological University	30	8.92	6.81	2.11
Amrita Vishwa Vidyapeetham	23	4.91	3.09	1.82
Indian Institute of Technology Roorkee	6	12.78	14.44	1.66

GMS (max 25)

Institute	Rank	Published	Predicted	error
Madan Mohan Malaviya University of Technolog	60	14.35	11.42	2.93
University of Hyderabad	74	14.51	12.18	2.33
Jamia Millia Islamia	24	15.64	13.69	1.95
National Institute of Technology Rourkela	13	16.93	15.17	1.76
National Institute of Technology Delhi	65	18.65	17.27	1.38

FPPP (max 10)

Institute	Rank	Published	Predicted	error
Indian Institute of Technology Patna	19	7.25	3.83	3.42
Indian Institute of Technology Bhilai	72	8.69	5.54	3.15
Indian Institute of Technology Jammu	56	6.36	3.29	3.07
Indian Institute of Technology Gandhinagar	25	8.61	5.69	2.92
Indian Institute of Technology Bombay	3	9.93	7.06	2.87

SS (max 20)

Institute	Rank	Published	Predicted	error
Amity University	37	13.6	16.47	2.87
Saveetha Institute of Medical and Technical	45	16.02	13.18	2.84
Indian Institute of Technology Mandi	26	12.45	9.79	2.66
Vignan's Foundation for Science, Technology	80	17.3	14.65	2.65
Chitkara University	89	13.34	15.47	2.13

FRU (max 30)

Institute	Rank	Published	Predicted	error
University of Hyderabad	74	14.38	22.859	8.479
Sant Longowal Institute of Engineering & Tec	79	22.29	17.824	4.466
Indian Institute of Engineering Science and	54	16.17	19.609	3.439
Banasthali Vidyapith	71	11.34	14.63	3.29
National Institute of Technology Meghalaya	83	17.72	20.935	3.215

The worst cases are not shared between parameters. Three institutes recur — University of Hyderabad, Madan Mohan Malaviya University of Technology and Jamia Millia Islamia — but the rest differ, and so do the failure modes.

The largest single error in the whole study is instructive. University of Hyderabad (rank 74) has its 2025 FRU over-predicted by 8.48 marks: 22.86 predicted against 14.38 published. The cause is upstream of the model. That institute reports a total student strength of 284 — the smallest in the 2025 top 90, against a median of 4,751 — while reporting operational expenditure within the normal range. Dividing the real expenditure by that strength gives per-student spending far above anything in the training years, and a log-linear fit extrapolates straight out of range. The extraction is faithful to the submitted document; the submitted number is the anomaly. A per-student ratio inherits whatever the denominator says, which is the structural weakness of the ESCS and FRU input definitions.

5.4 Why SS is the weak case

The prescribed SS inputs are both ratios, so they are invariant to how big the institute is. The published SS sub-score is not scale-free. On the 2023–2024 years alone, the Spearman correlation between the published SS sub-score and each quantity is:

Quantity	Spearman with published SS sub-score (2023–2024)
total_students	0.925
faculty_count	0.902
ss_nt	0.783
ss_x2	-0.088
ss_x1	0.083

The ratio inputs sit at zero; institute size sits at 0.93. A log-linear model on total student strength alone reaches LOIO CV MAE 1.08 (R^2 0.87) and 2025 MAE 1.03 (R^2 0.87, p 0.93) — far better than the prescribed inputs manage, which never beat a constant. That model is reported here as a diagnostic of the negative result; it is not the SS prediction in this report, which stays with the prescribed inputs and the model the CV selected (PreviousYear / per_year).

The practical conclusion is that a scale-free input cannot predict a scale-dependent sub-score. This is a limitation of the prescribed input set, not of the extraction or the fitting.

5.5 Combined score

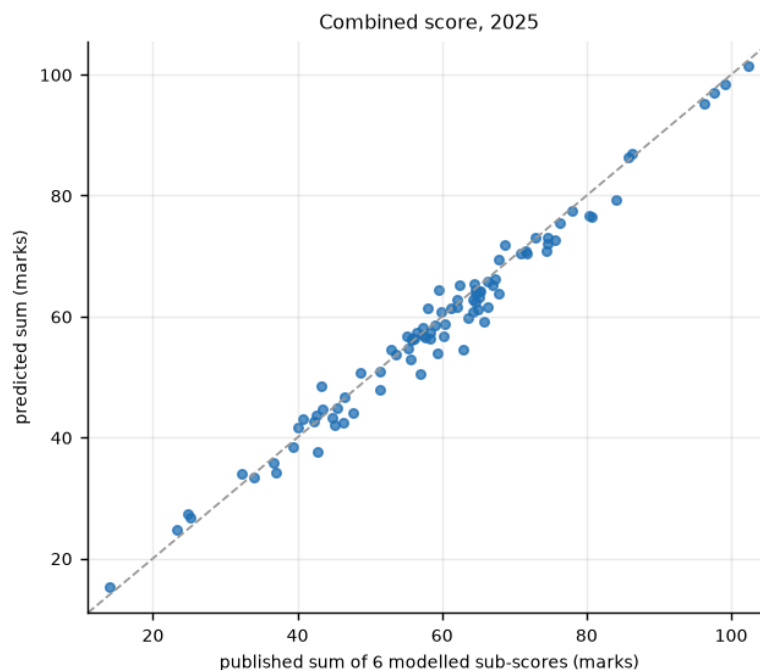


Figure 4 — the six modelled sub-scores summed per institute (125 marks).

Quantity	Value
institutes scored	90
MAE (marks of 125)	2.00
RMSE	2.63
R ²	0.974
bias	-0.92
within ±1 mark	42%
within ±2 marks	63%
Spearman	0.973
mean rank shift on this sum	4.02 of 90
largest rank shift	22

The sum is in marks and is deliberately not converted into points of the published normalised total: that needs NIRF's normalisation formula, which is not published, and guessing it would put a false precision on the result. Rank shifts look large in absolute terms but the sum has many ties — a great many institutes sit at a capped sub-score — so positions move freely while the underlying value barely changes.

6. Verdicts

Parameter	Verdict
ESCS	Predictable. Isotonic in the reimbursement percentage, pooled over both training years. Tight error and near-perfect rank agreement.
GPHD	Predictable, but the prescribed input is the weaker signal — the institute's own previous year wins on CV.
GMS	The most accurate of the six. The weighted-median-salary input reproduces the published sub-score closely, but last year's published value still edges.
FPPP	Weak, and the weakest of the six on CV. Sponsored and consultancy amounts per faculty are volatile year to year, so the previous-year baseline carries little signal.
SS	The prescribed ratio inputs carry no signal; see 5.4. The reported number is what the previous-year baseline gives, which is honest but is not a model.
FRU	The least accurate of the six (1.44 marks MAE) and the only one that misses badly on a single institute. Additive in log capital and operational spend.

7. What would make this better

IPR needs a source that exists. Nothing in this harness can predict it from the published PDFs.

Scale the SS inputs. Multiplicative features — admitted ÷ sanctioned intake × total strength, or the raw counts — would restore the signal the ratio form throws away. Section 5.4 quantifies what is left on the table (CV R² 0.87 available, ≈0 achieved).

Add a second training year of features. The previous-year baseline winning for four of six parameters is a symptom of thin training data, not of a strong prior: with 2022 features available the fitted families would get their first genuine advantage.

Read FPPP amounts more years out. Three financial years of a volatile quantity is thin; the volatility, not the extraction, is what limits it.

Appendix A — prescribed inputs against the published sub-score, 2023+2024

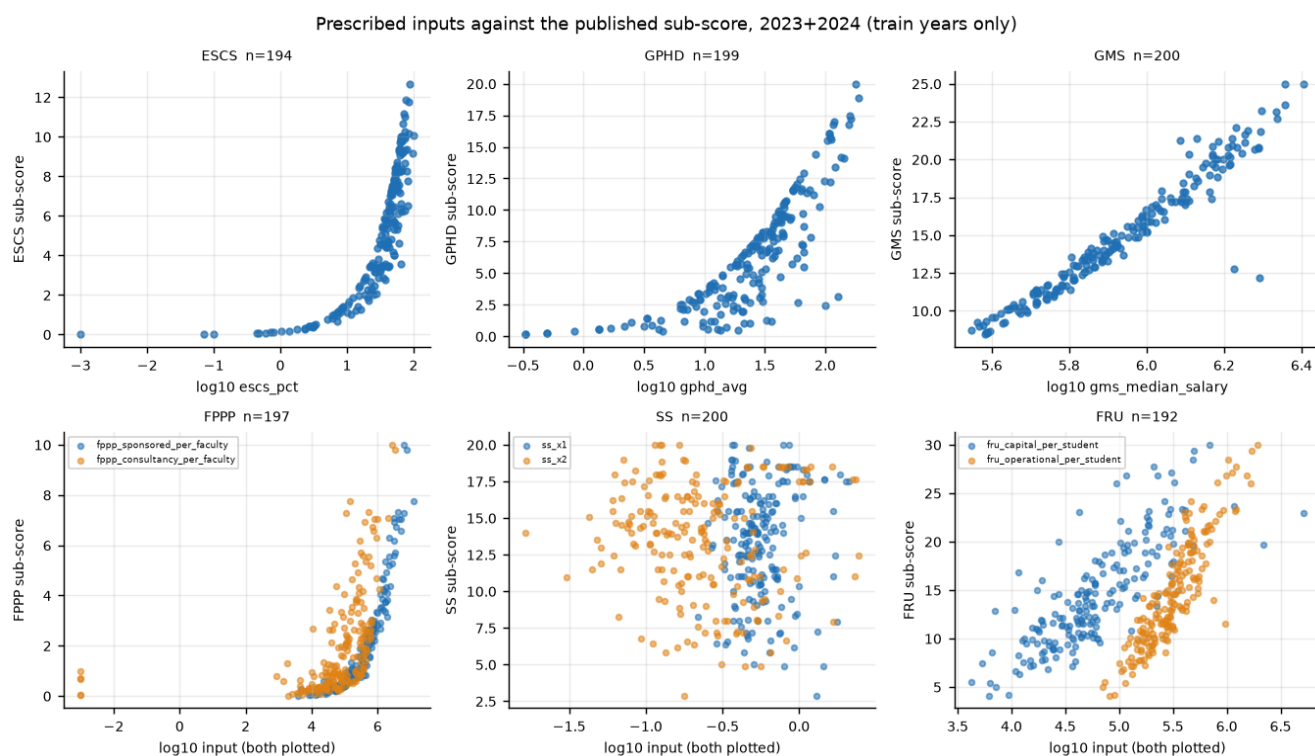


Figure 5 — the prescribed inputs, on training years only. For the two-input parameters both inputs are overlaid in different colours. ESCS, GMS and FRU show the expected monotone structure; SS shows the flatness quantified in 5.4.

Appendix B — full 2025 predictions

ESCS — 88 scored institutes

Rank	Institute	Published	Predicted	Error
1	Indian Institute of Technology Madras	7.66	6.88	-0.78
2	Indian Institute of Technology Delhi	7.25	6.45	-0.80
3	Indian Institute of Technology Bombay	5.37	6.33	+0.96
4	Indian Institute of Technology Kanpur	7.21	7.57	+0.36
5	Indian Institute of Technology Kharagpur	7.80	6.45	-1.35
6	Indian Institute of Technology Roorkee	7.37	6.45	-0.92
7	Indian Institute of Technology Hyderabad	4.59	4.06	-0.53
8	Indian Institute of Technology Guwahati	6.08	4.71	-1.37
9	National Institute of Technology Tiruchirappal	8.21	7.57	-0.64
10	Indian Institute of Technology (Banaras Hindu	10.44	9.74	-0.70
11	Birla Institute of Technology & Science -Pilan	0.13	0.13	+0.00
12	Indian Institute of Technology Indore	6.85	6.33	-0.52
13	National Institute of Technology Rourkela	9.13	7.58	-1.55
14	S.R.M. Institute of Science and Technology	1.10	0.97	-0.13
15	Indian Institute of Technology (Indian School	7.91	7.57	-0.34
16	Vellore Institute of Technology	0.12	0.12	+0.00
17	National Institute of Technology Karnataka, Su	9.55	8.35	-1.20
18	Jadavpur University	5.69	7.58	+1.89
19	Indian Institute of Technology Patna	9.57	9.74	+0.17
20	Anna University	5.46	5.51	+0.05
21	National Institute of Technology Calicut	10.34	9.74	-0.60
22	Siksha 'O' Anusandhan	1.40	1.28	-0.12

Rank	Institute	Published	Predicted	Error
23	Amrita Vishwa Vidyapeetham	3.37	2.44	-0.93
24	Jamia Millia Islamia	9.02	11.23	+2.21
25	Indian Institute of Technology Gandhinagar	4.10	4.70	+0.60
26	Indian Institute of Technology Mandi	5.67	6.45	+0.78
27	Indian Institute of Technology Jodhpur	6.21	6.33	+0.12
28	National Institute of Technology Warangal	7.52	6.88	-0.64
29	Thapar Institute of Engineering and Technology	1.96	1.79	-0.17
30	Delhi Technological University	1.77	1.79	+0.02
31	Chandigarh University	1.98	1.79	-0.19
32	Indian Institute of Technology Ropar	7.61	6.45	-1.16
33	Kalasalingam Academy of Research and Education	7.68	6.45	-1.23
34	Aligarh Muslim University	3.59	4.22	+0.63
35	Koneru Lakshmaiah Education Foundation Univers	1.40	1.28	-0.12
36	Kalinga Institute of Industrial Technology	2.06	1.79	-0.27
37	Amity University	0.23	0.23	-0.00
38	International Institute of Information Technol	2.66	2.44	-0.22
39	Indian Institute of Technology Bhubaneswar	5.34	6.45	+1.11
40	Shanmugha Arts Science Technology & Research A	5.08	4.06	-1.02
41	Institute of Chemical Technology	3.06	4.06	+1.00
42	Malaviya National Institute of Technology	7.27	6.45	-0.82
43	UPES	0.12	0.12	+0.00
44	Visvesvaraya National Institute of Technology,	5.38	6.45	+1.07
45	Saveetha Institute of Medical and Technical Sc	4.12	3.26	-0.86
46	Symbiosis International	0.51	0.52	+0.01
47	Sri Sivasubramaniya Nadar College of Engineeri	8.04	7.58	-0.46
48	Lovely Professional University	2.43	2.05	-0.38
49	National Institute of Technology Durgapur	9.22	8.35	-0.87
50	National Institute of Technology Silchar	6.11	6.45	+0.34
51	Birla Institute of Technology	1.71	2.05	+0.34
52	Graphic Era University	0.96	0.96	+0.00
53	National Institute of Technology Patna	11.90	11.23	-0.67
54	Indian Institute of Engineering Science and Te	6.50	7.57	+1.07
55	Dr. B R Ambedkar National Institute of Technol	7.82	6.45	-1.37
56	Indian Institute of Technology Jammu	5.73	5.51	-0.22
57	Indian Institute of Technology, Tirupati	7.25	7.57	+0.32
58	Manipal University Jaipur	0.15	0.15	+0.00
59	Manipal Institute of Technology	0.70	0.70	+0.00
60	Madan Mohan Malaviya University of Technology	10.28	9.74	-0.54
61	Indian Institute of Space Science and Technolo	5.86	6.45	+0.59
62	Motilal Nehru National Institute of Technology	9.96	9.74	-0.22
63	Indraprastha Institute of Information Technolo	0.99	1.28	+0.29
64	Indian Institute of Technology Palakkad	5.53	6.88	+1.35
65	National Institute of Technology Delhi	11.08	9.74	-1.34
66	Sardar Vallabhbhai National Institute of Techn	3.86	4.70	+0.84
67	PSG College of Technology	2.07	2.44	+0.37
67	Sathyabama Institute of Science and Technology	2.07	1.79	-0.28

Rank	Institute	Published	Predicted	Error
70	Netaji Subhas University of Technology (NSUT)	3.34	3.34	+0.00
71	Banasthali Vidyapith	3.79	4.06	+0.27
72	Indian Institute of Technology Bhilai	6.85	6.33	-0.52
73	National Institute of Technology Srinagar	13.80	11.23	-2.57
75	M. S. Ramaiah Institute of Technology	2.39	3.34	+0.95
76	Christ University	2.22	2.05	-0.17
77	Indian Institute of Technology Dharwad	4.93	6.45	+1.52
78	Rajiv Gandhi Institute of Petroleum Technology	1.90	2.05	+0.15
79	Sant Longowal Institute of Engineering & Techn	2.56	2.86	+0.30
80	Vignan's Foundation for Science, Technology an	7.28	6.45	-0.83
81	Maulana Azad National Institute of Technology	5.97	8.35	+2.38
82	National Institute of Technology, Jamshedpur	10.51	9.74	-0.77
83	National Institute of Technology Meghalaya	11.22	9.74	-1.48
84	Jain university,Bangalore	2.27	3.26	+0.99
85	National Institute of Technology Kurukshetra	8.73	7.58	-1.15
86	National Institute of Technology, Raipur	8.51	7.57	-0.94
87	Vel Tech Rangarajan Dr. Sagunthala R & D Insti	4.34	3.26	-1.08
88	AU College of Engineering (A)	11.46	11.23	-0.23
89	Chitkara University	0.95	0.97	+0.02
90	COEP Technological University	5.51	7.57	+2.06

GPHD — 86 scored institutes

Rank	Institute	Published	Predicted	Error
1	Indian Institute of Technology Madras	16.68	16.98	+0.30
2	Indian Institute of Technology Delhi	17.89	17.49	-0.40
3	Indian Institute of Technology Bombay	20.00	18.87	-1.13
4	Indian Institute of Technology Kanpur	15.30	16.08	+0.78
5	Indian Institute of Technology Kharagpur	19.79	20.00	+0.21
6	Indian Institute of Technology Roorkee	12.78	14.44	+1.66
7	Indian Institute of Technology Hyderabad	8.69	9.01	+0.32
8	Indian Institute of Technology Guwahati	15.50	15.71	+0.21
9	National Institute of Technology Tiruchirappal	10.69	11.35	+0.66
10	Indian Institute of Technology (Banaras Hindu	13.76	12.90	-0.86
11	Birla Institute of Technology & Science -Pilan	7.58	7.51	-0.07
12	Indian Institute of Technology Indore	9.23	9.96	+0.73
13	National Institute of Technology Rourkela	12.01	11.63	-0.38
14	S.R.M. Institute of Science and Technology	11.65	10.29	-1.36
15	Indian Institute of Technology (Indian School	14.57	14.18	-0.39
16	Vellore Institute of Technology	13.72	12.31	-1.41
17	National Institute of Technology Karnataka, Su	10.78	11.70	+0.92
18	Jadavpur University	11.91	12.46	+0.55
19	Indian Institute of Technology Patna	9.93	8.80	-1.13
20	Anna University	12.66	13.43	+0.77
21	National Institute of Technology Calicut	9.66	10.62	+0.96
22	Siksha `O` Anusandhan	10.61	9.91	-0.70
23	Amrita Vishwa Vidyapeetham	4.91	3.09	-1.82
24	Jamia Millia Islamia	7.90	8.26	+0.36
25	Indian Institute of Technology Gandhinagar	6.58	7.66	+1.08
26	Indian Institute of Technology Mandi	6.94	7.85	+0.91
27	Indian Institute of Technology Jodhpur	3.14	3.56	+0.42
28	National Institute of Technology Warangal	9.06	9.50	+0.44
29	Thapar Institute of Engineering and Technology	9.39	9.45	+0.06
30	Delhi Technological University	8.92	6.81	-2.11
31	Chandigarh University	8.40	9.65	+1.25
32	Indian Institute of Technology Ropar	8.40	6.85	-1.55
33	Kalasalingam Academy of Research and Education	9.08	9.22	+0.14
34	Aligarh Muslim University	5.72	6.24	+0.52
35	Koneru Lakshmaiah Education Foundation Univers	11.07	9.72	-1.35
36	Kalinga Institute of Industrial Technology	12.05	11.82	-0.23
37	Amity University	6.73	6.67	-0.06
38	International Institute of Information Technol	2.81	2.71	-0.10
39	Indian Institute of Technology Bhubaneswar	6.99	6.64	-0.35
40	Shanmugha Arts Science Technology & Research A	10.65	9.67	-0.98
41	Institute of Chemical Technology	6.88	9.04	+2.16
42	Malaviya National Institute of Technology	8.30	8.23	-0.07
43	UPES	0.39	0.60	+0.21
44	Visvesvaraya National Institute of Technology,	7.84	7.79	-0.05
45	Saveetha Institute of Medical and Technical Sc	3.09	2.06	-1.03

Rank	Institute	Published	Predicted	Error
47	Sri Sivasubramaniya Nadar College of Engineeri	5.98	6.71	+0.73
48	Lovely Professional University	4.91	4.75	-0.16
49	National Institute of Technology Durgapur	8.78	9.45	+0.67
50	National Institute of Technology Silchar	7.82	8.18	+0.36
51	Birla Institute of Technology	5.58	5.43	-0.15
52	Graphic Era University	2.22	3.09	+0.87
53	National Institute of Technology Patna	7.84	6.95	-0.89
54	Indian Institute of Engineering Science and Te	9.30	10.69	+1.39
55	Dr. B R Ambedkar National Institute of Technol	6.27	6.57	+0.30
56	Indian Institute of Technology Jammu	2.06	1.43	-0.63
57	Indian Institute of Technology, Tirupati	0.77	0.60	-0.17
58	Manipal University Jaipur	1.01	1.16	+0.15
59	Manipal Institute of Technology	5.13	4.79	-0.34
60	Madan Mohan Malaviya University of Technology	7.90	3.86	-4.04
61	Indian Institute of Space Science and Technolo	4.01	3.72	-0.29
62	Motilal Nehru National Institute of Technology	6.70	7.94	+1.24
63	Indraprastha Institute of Information Technolo	3.96	4.01	+0.05
64	Indian Institute of Technology Palakkad	1.52	1.10	-0.42
65	National Institute of Technology Delhi	3.18	2.76	-0.42
66	Sardar Vallabhbhai National Institute of Techn	6.14	6.88	+0.74
67	PSG College of Technology	2.71	3.31	+0.60
67	Sathyabama Institute of Science and Technology	4.79	6.17	+1.38
69	International Institute of Information Technol	1.47	1.30	-0.17
70	Netaji Subhas University of Technology (NSUT)	6.73	7.22	+0.49
71	Banasthali Vidyapith	7.33	7.82	+0.49
72	Indian Institute of Technology Bhilai	0.83	0.67	-0.16
73	National Institute of Technology Srinagar	7.44	7.22	-0.22
74	University of Hyderabad	5.51	5.94	+0.43
75	M. S. Ramaiah Institute of Technology	1.96	1.94	-0.02
76	Christ University	0.45	0.82	+0.37
78	Rajiv Gandhi Institute of Petroleum Technology	2.81	2.93	+0.12
79	Sant Longowal Institute of Engineering & Techn	2.62	3.09	+0.47
80	Vignan's Foundation for Science, Technology an	5.41	4.52	-0.89
81	Maulana Azad National Institute of Technology	5.68	5.34	-0.34
83	National Institute of Technology Meghalaya	3.23	3.36	+0.13
84	Jain university,Bangalore	3.49	4.06	+0.57
85	National Institute of Technology Kurukshetra	4.45	4.88	+0.43
86	National Institute of Technology, Raipur	7.31	7.76	+0.45
87	Vel Tech Rangarajan Dr. Sagunthala R & D Insti	1.41	1.03	-0.38
88	AU College of Engineering (A)	9.08	8.71	-0.37
89	Chitkara University	3.49	3.20	-0.29

GMS — 86 scored institutes

Rank	Institute	Published	Predicted	Error
1	Indian Institute of Technology Madras	19.54	19.34	-0.20
2	Indian Institute of Technology Delhi	20.91	20.93	+0.02
3	Indian Institute of Technology Bombay	20.67	19.99	-0.68
4	Indian Institute of Technology Kanpur	21.62	21.84	+0.22
5	Indian Institute of Technology Kharagpur	21.34	20.87	-0.47
6	Indian Institute of Technology Roorkee	19.98	20.17	+0.19
7	Indian Institute of Technology Hyderabad	21.27	20.79	-0.48
8	Indian Institute of Technology Guwahati	21.50	21.63	+0.13
9	National Institute of Technology Tiruchirappal	17.86	17.34	-0.52
10	Indian Institute of Technology (Banaras Hindu	20.81	20.43	-0.38
11	Birla Institute of Technology & Science -Pilan	20.22	19.62	-0.60
12	Indian Institute of Technology Indore	20.85	20.22	-0.63
13	National Institute of Technology Rourkela	16.93	15.17	-1.76
14	S.R.M. Institute of Science and Technology	10.78	10.03	-0.75
15	Indian Institute of Technology (Indian School	17.84	17.86	+0.02
16	Vellore Institute of Technology	13.93	13.37	-0.56
17	National Institute of Technology Karnataka, Su	17.80	17.18	-0.62
18	Jadavpur University	15.19	15.20	+0.01
19	Indian Institute of Technology Patna	21.20	20.74	-0.46
20	Anna University	12.61	11.74	-0.87
21	National Institute of Technology Calicut	16.30	15.84	-0.46
22	Siksha `O` Anusandhan	10.56	9.88	-0.68
23	Amrita Vishwa Vidyapeetham	11.81	10.72	-1.09
24	Jamia Millia Islamia	15.64	13.69	-1.95
25	Indian Institute of Technology Gandhinagar	18.27	17.47	-0.80
26	Indian Institute of Technology Mandi	21.20	21.37	+0.17
27	Indian Institute of Technology Jodhpur	18.04	18.27	+0.23
28	National Institute of Technology Warangal	18.45	17.84	-0.61
29	Thapar Institute of Engineering and Technology	15.55	15.24	-0.31
30	Delhi Technological University	17.49	17.24	-0.25
31	Chandigarh University	13.17	12.91	-0.26
32	Indian Institute of Technology Ropar	20.05	19.50	-0.55
33	Kalasalingam Academy of Research and Education	11.39	11.07	-0.32
34	Aligarh Muslim University	11.06	11.09	+0.03
35	Koneru Lakshmaiah Education Foundation Univers	13.06	12.06	-1.00
36	Kalinga Institute of Industrial Technology	11.77	11.63	-0.14
37	Amity University	10.78	9.95	-0.83
38	International Institute of Information Technol	25.00	25.00	+0.00
39	Indian Institute of Technology Bhubaneswar	18.42	18.56	+0.14
40	Shanmugha Arts Science Technology & Research A	12.91	12.57	-0.34
41	Institute of Chemical Technology	13.64	13.06	-0.58
42	Malaviya National Institute of Technology	15.45	14.89	-0.56
43	UPES	11.25	10.64	-0.61
44	Visvesvaraya National Institute of Technology,	15.55	15.62	+0.07
45	Saveetha Institute of Medical and Technical Sc	9.04	8.61	-0.43

Rank	Institute	Published	Predicted	Error
47	Sri Sivasubramaniya Nadar College of Engineeri	13.39	12.57	-0.82
48	Lovely Professional University	12.54	11.47	-1.07
49	National Institute of Technology Durgapur	15.83	14.89	-0.94
50	National Institute of Technology Silchar	16.53	15.91	-0.62
51	Birla Institute of Technology	15.06	14.88	-0.18
52	Graphic Era University	10.95	11.31	+0.36
53	National Institute of Technology Patna	14.19	14.34	+0.15
54	Indian Institute of Engineering Science and Te	13.42	13.21	-0.21
55	Dr. B R Ambedkar National Institute of Technol	13.89	13.66	-0.23
56	Indian Institute of Technology Jammu	18.17	17.86	-0.31
57	Indian Institute of Technology, Tirupati	17.23	17.01	-0.22
58	Manipal University Jaipur	12.13	11.24	-0.89
59	Manipal Institute of Technology	13.75	13.33	-0.42
60	Madan Mohan Malaviya University of Technology	14.35	11.42	-2.93
61	Indian Institute of Space Science and Technolo	15.90	15.87	-0.03
62	Motilal Nehru National Institute of Technology	17.63	17.22	-0.41
63	Indraprastha Institute of Information Technolo	19.95	19.67	-0.28
64	Indian Institute of Technology Palakkad	17.01	16.55	-0.46
65	National Institute of Technology Delhi	18.65	17.27	-1.38
66	Sardar Vallabhbhai National Institute of Techn	14.28	13.66	-0.62
67	PSG College of Technology	12.17	11.63	-0.54
67	Sathyabama Institute of Science and Technology	10.07	9.15	-0.92
69	International Institute of Information Technol	23.58	22.68	-0.90
70	Netaji Subhas University of Technology (NSUT)	19.26	18.85	-0.41
71	Banasthali Vidyapith	13.95	12.93	-1.02
72	Indian Institute of Technology Bhilai	17.63	17.43	-0.20
73	National Institute of Technology Srinagar	15.14	14.02	-1.12
74	University of Hyderabad	14.51	12.18	-2.33
75	M. S. Ramaiah Institute of Technology	12.56	12.06	-0.50
76	Christ University	10.11	9.84	-0.27
78	Rajiv Gandhi Institute of Petroleum Technology	13.79	13.69	-0.10
79	Sant Longowal Institute of Engineering & Techn	8.66	8.57	-0.09
80	Vignan's Foundation for Science, Technology an	9.68	10.39	+0.71
81	Maulana Azad National Institute of Technology	16.50	16.00	-0.50
83	National Institute of Technology Meghalaya	15.04	14.13	-0.91
84	Jain university,Bangalore	11.95	10.96	-0.99
85	National Institute of Technology Kurukshetra	15.34	15.55	+0.21
86	National Institute of Technology, Raipur	14.23	14.17	-0.06
87	Vel Tech Rangarajan Dr. Sagunthala R & D Insti	8.58	8.53	-0.05
88	AU College of Engineering (A)	11.49	11.39	-0.10
89	Chitkara University	12.11	12.02	-0.09

FPPP — 86 scored institutes

Rank	Institute	Published	Predicted	Error
1	Indian Institute of Technology Madras	10.00	9.78	-0.22
2	Indian Institute of Technology Delhi	9.01	6.56	-2.45
3	Indian Institute of Technology Bombay	9.93	7.06	-2.87
4	Indian Institute of Technology Kanpur	8.85	6.72	-2.13
5	Indian Institute of Technology Kharagpur	6.18	3.75	-2.43
6	Indian Institute of Technology Roorkee	6.36	4.25	-2.11
7	Indian Institute of Technology Hyderabad	7.68	5.18	-2.50
8	Indian Institute of Technology Guwahati	7.05	5.58	-1.47
9	National Institute of Technology Tiruchirappal	2.34	1.55	-0.79
10	Indian Institute of Technology (Banaras Hindu	3.74	2.19	-1.55
11	Birla Institute of Technology & Science -Pilan	1.63	0.81	-0.82
12	Indian Institute of Technology Indore	5.93	3.56	-2.37
13	National Institute of Technology Rourkela	3.74	2.01	-1.73
14	S.R.M. Institute of Science and Technology	0.50	0.26	-0.24
15	Indian Institute of Technology (Indian School	3.52	1.93	-1.59
16	Vellore Institute of Technology	0.42	0.20	-0.22
17	National Institute of Technology Karnataka, Su	4.14	2.93	-1.21
18	Jadavpur University	3.68	2.66	-1.02
19	Indian Institute of Technology Patna	7.25	3.83	-3.42
20	Anna University	3.21	1.97	-1.24
21	National Institute of Technology Calicut	1.69	1.09	-0.60
22	Siksha `O` Anusandhan	1.57	0.89	-0.68
23	Amrita Vishwa Vidyapeetham	3.18	2.13	-1.05
24	Jamia Millia Islamia	4.93	2.92	-2.01
25	Indian Institute of Technology Gandhinagar	8.61	5.69	-2.92
26	Indian Institute of Technology Mandi	3.74	2.11	-1.63
27	Indian Institute of Technology Jodhpur	6.41	4.17	-2.24
28	National Institute of Technology Warangal	2.05	0.99	-1.06
29	Thapar Institute of Engineering and Technology	1.01	0.58	-0.43
30	Delhi Technological University	1.14	0.64	-0.50
31	Chandigarh University	0.82	0.32	-0.50
32	Indian Institute of Technology Ropar	6.10	4.11	-1.99
33	Kalasalingam Academy of Research and Education	0.85	0.53	-0.32
34	Aligarh Muslim University	2.46	2.03	-0.43
35	Koneru Lakshmaiah Education Foundation Univers	0.74	0.36	-0.38
36	Kalinga Institute of Industrial Technology	2.75	0.64	-2.11
37	Amity University	0.87	0.65	-0.22
38	International Institute of Information Technol	6.52	4.37	-2.15
39	Indian Institute of Technology Bhubaneswar	4.21	2.27	-1.94
40	Shanmugha Arts Science Technology & Research A	3.37	2.31	-1.06
41	Institute of Chemical Technology	6.42	4.31	-2.11
42	Malaviya National Institute of Technology	3.85	1.70	-2.15
43	UPES	0.54	0.28	-0.26
44	Visvesvaraya National Institute of Technology,	3.66	2.39	-1.27
45	Saveetha Institute of Medical and Technical Sc	0.79	0.20	-0.59

Rank	Institute	Published	Predicted	Error
47	Sri Sivasubramaniya Nadar College of Engineeri	1.41	0.75	-0.66
48	Lovely Professional University	0.21	0.08	-0.13
49	National Institute of Technology Durgapur	2.74	1.59	-1.15
50	National Institute of Technology Silchar	2.11	1.22	-0.89
51	Birla Institute of Technology	2.10	1.04	-1.06
52	Graphic Era University	1.14	0.59	-0.55
53	National Institute of Technology Patna	2.28	1.21	-1.07
54	Indian Institute of Engineering Science and Te	1.86	1.07	-0.79
55	Dr. B R Ambedkar National Institute of Technol	0.97	0.27	-0.70
56	Indian Institute of Technology Jammu	6.36	3.29	-3.07
57	Indian Institute of Technology, Tirupati	4.63	2.60	-2.03
58	Manipal University Jaipur	0.77	0.40	-0.37
59	Manipal Institute of Technology	0.28	0.12	-0.16
60	Madan Mohan Malaviya University of Technology	3.17	0.80	-2.37
61	Indian Institute of Space Science and Technolo	2.23	0.92	-1.31
62	Motilal Nehru National Institute of Technology	2.01	1.40	-0.61
63	Indraprastha Institute of Information Technolo	4.54	3.16	-1.38
64	Indian Institute of Technology Palakkad	4.69	2.99	-1.70
65	National Institute of Technology Delhi	2.11	0.96	-1.15
66	Sardar Vallabhbhai National Institute of Techn	2.45	1.05	-1.40
67	PSG College of Technology	0.70	0.26	-0.44
67	Sathyabama Institute of Science and Technology	1.07	0.56	-0.51
69	International Institute of Information Technol	8.13	7.77	-0.36
70	Netaji Subhas University of Technology (NSUT)	0.41	0.35	-0.06
71	Banasthali Vidyapith	1.37	0.65	-0.72
72	Indian Institute of Technology Bhilai	8.69	5.54	-3.15
73	National Institute of Technology Srinagar	2.11	1.07	-1.04
74	University of Hyderabad	4.29	2.66	-1.63
75	M. S. Ramaiah Institute of Technology	0.59	0.30	-0.29
76	Christ University	0.14	0.06	-0.08
78	Rajiv Gandhi Institute of Petroleum Technology	3.20	1.95	-1.25
79	Sant Longowal Institute of Engineering & Techn	0.15	0.10	-0.05
80	Vignan's Foundation for Science, Technology an	0.89	0.35	-0.54
81	Maulana Azad National Institute of Technology	1.87	0.81	-1.06
83	National Institute of Technology Meghalaya	3.69	2.15	-1.54
84	Jain university,Bangalore	1.27	0.54	-0.73
85	National Institute of Technology Kurukshetra	1.76	0.82	-0.94
86	National Institute of Technology, Raipur	1.07	0.50	-0.57
87	Vel Tech Rangarajan Dr. Sagunthala R & D Insti	0.52	0.22	-0.30
88	AU College of Engineering (A)	1.73	1.58	-0.15
89	Chitkara University	0.13	0.03	-0.10

SS — 86 scored institutes

Rank	Institute	Published	Predicted	Error
1	Indian Institute of Technology Madras	18.50	18.50	+0.00
2	Indian Institute of Technology Delhi	16.45	18.50	+2.05
3	Indian Institute of Technology Bombay	18.50	18.50	+0.00
4	Indian Institute of Technology Kanpur	18.18	18.33	+0.15
5	Indian Institute of Technology Kharagpur	18.23	19.76	+1.53
6	Indian Institute of Technology Roorkee	17.75	17.66	-0.09
7	Indian Institute of Technology Hyderabad	12.49	12.41	-0.08
8	Indian Institute of Technology Guwahati	17.86	18.32	+0.46
9	National Institute of Technology Tiruchirappal	17.50	17.50	+0.00
10	Indian Institute of Technology (Banaras Hindu	17.62	16.28	-1.34
11	Birla Institute of Technology & Science -Pilan	19.00	17.87	-1.13
12	Indian Institute of Technology Indore	12.55	10.73	-1.82
13	National Institute of Technology Rourkela	17.50	17.50	+0.00
14	S.R.M. Institute of Science and Technology	17.82	18.04	+0.22
15	Indian Institute of Technology (Indian School	15.87	15.43	-0.44
16	Vellore Institute of Technology	20.00	20.00	+0.00
17	National Institute of Technology Karnataka, Su	17.50	17.50	+0.00
18	Jadavpur University	18.25	18.50	+0.25
19	Indian Institute of Technology Patna	12.48	12.02	-0.46
20	Anna University	17.86	15.94	-1.92
21	National Institute of Technology Calicut	16.00	16.00	+0.00
22	Siksha `O` Anusandhan	19.00	17.50	-1.50
23	Amrita Vishwa Vidyapeetham	18.00	16.22	-1.78
24	Jamia Millia Islamia	10.50	10.50	+0.00
25	Indian Institute of Technology Gandhinagar	11.50	11.50	+0.00
26	Indian Institute of Technology Mandi	12.45	9.79	-2.66
27	Indian Institute of Technology Jodhpur	12.50	12.53	+0.03
28	National Institute of Technology Warangal	16.16	16.50	+0.34
29	Thapar Institute of Engineering and Technology	18.00	18.23	+0.23
30	Delhi Technological University	19.00	19.00	+0.00
31	Chandigarh University	17.88	17.95	+0.07
32	Indian Institute of Technology Ropar	11.70	12.41	+0.71
33	Kalasalingam Academy of Research and Education	16.49	16.48	-0.01
34	Aligarh Muslim University	11.00	11.00	+0.00
35	Koneru Lakshmaiah Education Foundation Univers	18.65	17.47	-1.18
36	Kalinga Institute of Industrial Technology	20.00	20.00	+0.00
37	Amity University	13.60	16.47	+2.87
38	International Institute of Information Technol	9.08	9.33	+0.25
39	Indian Institute of Technology Bhubaneswar	11.05	11.15	+0.10
40	Shanmugha Arts Science Technology & Research A	17.50	16.32	-1.18
41	Institute of Chemical Technology	11.50	10.37	-1.13
42	Malaviya National Institute of Technology	15.39	14.97	-0.42
43	UPES	15.50	15.50	+0.00
44	Visvesvaraya National Institute of Technology,	13.31	13.94	+0.63
45	Saveetha Institute of Medical and Technical Sc	16.02	13.18	-2.84

Rank	Institute	Published	Predicted	Error
47	Sri Sivasubramaniya Nadar College of Engineeri	12.50	13.50	+1.00
48	Lovely Professional University	18.00	18.00	+0.00
49	National Institute of Technology Durgapur	14.07	14.09	+0.02
50	National Institute of Technology Silchar	14.42	14.35	-0.07
51	Birla Institute of Technology	13.45	13.49	+0.04
52	Graphic Era University	12.75	12.70	-0.05
53	National Institute of Technology Patna	12.30	12.35	+0.05
54	Indian Institute of Engineering Science and Te	12.27	12.26	-0.01
55	Dr. B R Ambedkar National Institute of Technol	14.00	14.10	+0.10
56	Indian Institute of Technology Jammu	7.95	7.98	+0.03
57	Indian Institute of Technology, Tirupati	8.71	8.32	-0.39
58	Manipal University Jaipur	14.84	14.67	-0.17
59	Manipal Institute of Technology	15.89	15.12	-0.77
60	Madan Mohan Malaviya University of Technology	13.50	13.22	-0.28
61	Indian Institute of Space Science and Technolo	8.65	8.95	+0.30
62	Motilal Nehru National Institute of Technology	15.79	16.01	+0.22
63	Indraprastha Institute of Information Technolo	12.00	11.70	-0.30
64	Indian Institute of Technology Palakkad	7.23	7.17	-0.06
65	National Institute of Technology Delhi	8.50	7.96	-0.54
66	Sardar Vallabhbhai National Institute of Techn	13.20	12.96	-0.24
67	PSG College of Technology	15.19	14.62	-0.57
67	Sathyabama Institute of Science and Technology	17.99	16.53	-1.46
69	International Institute of Information Technol	8.19	8.43	+0.24
70	Netaji Subhas University of Technology (NSUT)	15.48	15.04	-0.44
71	Banasthali Vidyapith	10.89	10.68	-0.21
72	Indian Institute of Technology Bhilai	6.22	6.36	+0.14
73	National Institute of Technology Srinagar	13.01	12.79	-0.22
74	University of Hyderabad	4.61	4.90	+0.29
75	M. S. Ramaiah Institute of Technology	14.50	14.40	-0.10
76	Christ University	10.15	8.26	-1.89
78	Rajiv Gandhi Institute of Petroleum Technology	9.50	8.00	-1.50
79	Sant Longowal Institute of Engineering & Techn	10.00	10.00	+0.00
80	Vignan's Foundation for Science, Technology an	17.30	14.65	-2.65
81	Maulana Azad National Institute of Technology	14.72	15.01	+0.29
83	National Institute of Technology Meghalaya	6.44	6.64	+0.20
84	Jain university,Bangalore	14.05	13.23	-0.82
85	National Institute of Technology Kurukshetra	15.19	14.63	-0.56
86	National Institute of Technology, Raipur	13.82	13.94	+0.12
87	Vel Tech Rangarajan Dr. Sagunthala R & D Insti	15.50	15.13	-0.37
88	AU College of Engineering (A)	11.50	11.49	-0.01
89	Chitkara University	13.34	15.47	+2.13

FRU — 85 scored institutes

Rank	Institute	Published	Predicted	Error
1	Indian Institute of Technology Madras	30.00	30.00	+0.00
2	Indian Institute of Technology Delhi	27.72	28.44	+0.72
4	Indian Institute of Technology Kanpur	26.55	26.36	-0.19
5	Indian Institute of Technology Kharagpur	22.88	24.42	+1.54
6	Indian Institute of Technology Roorkee	21.46	23.40	+1.94
7	Indian Institute of Technology Hyderabad	25.71	25.13	-0.58
8	Indian Institute of Technology Guwahati	18.32	21.01	+2.69
9	National Institute of Technology Tiruchirappal	19.73	20.24	+0.51
11	Birla Institute of Technology & Science -Pilani	18.40	19.17	+0.77
12	Indian Institute of Technology Indore	20.19	21.85	+1.66
14	S.R.M. Institute of Science and Technology	13.48	15.05	+1.57
15	Indian Institute of Technology (Indian School	18.20	20.50	+2.30
16	Vellore Institute of Technology	10.20	10.33	+0.13
17	National Institute of Technology Karnataka, Su	24.24	21.60	-2.64
18	Jadavpur University	13.97	15.53	+1.56
19	Indian Institute of Technology Patna	20.20	21.27	+1.07
20	Anna University	12.80	15.27	+2.47
21	National Institute of Technology Calicut	18.89	19.81	+0.92
22	Siksha 'O' Anusandhan	17.06	17.24	+0.18
23	Amrita Vishwa Vidyapeetham	15.73	15.98	+0.25
24	Jamia Millia Islamia	16.67	17.72	+1.05
25	Indian Institute of Technology Gandhinagar	25.57	24.93	-0.64
26	Indian Institute of Technology Mandi	21.51	23.27	+1.76
27	Indian Institute of Technology Jodhpur	25.44	25.50	+0.06
28	National Institute of Technology Warangal	17.68	18.76	+1.08
29	Thapar Institute of Engineering and Technology	15.31	16.10	+0.79
30	Delhi Technological University	12.00	13.19	+1.19
31	Chandigarh University	10.77	11.83	+1.06
32	Indian Institute of Technology Ropar	20.73	23.68	+2.95
33	Kalasalingam Academy of Research and Education	12.78	13.65	+0.87
34	Aligarh Muslim University	17.60	16.35	-1.25
35	Koneru Lakshmaiah Education Foundation Univers	10.66	11.99	+1.33
36	Kalinga Institute of Industrial Technology	15.61	16.83	+1.22
38	International Institute of Information Technol	19.20	20.26	+1.06
39	Indian Institute of Technology Bhubaneswar	18.49	20.37	+1.88
40	Shanmugha Arts Science Technology & Research A	14.07	14.84	+0.77
41	Institute of Chemical Technology	18.30	19.84	+1.54
42	Malaviya National Institute of Technology	14.90	17.02	+2.12
43	UPES	16.99	16.21	-0.78
44	Visvesvaraya National Institute of Technology,	16.73	19.05	+2.32
45	Saveetha Institute of Medical and Technical Sc	9.76	10.32	+0.56
46	Symbiosis International	13.50	14.78	+1.28
47	Sri Sivasubramaniya Nadar College of Engineeri	14.50	15.20	+0.70
48	Lovely Professional University	7.08	5.66	-1.42
49	National Institute of Technology Durgapur	15.67	17.42	+1.75

Rank	Institute	Published	Predicted	Error
50	National Institute of Technology Silchar	15.06	16.61	+1.55
51	Birla Institute of Technology	15.72	16.90	+1.18
52	Graphic Era University	11.97	12.98	+1.01
53	National Institute of Technology Patna	13.50	15.45	+1.95
54	Indian Institute of Engineering Science and Te	16.17	19.61	+3.44
55	Dr. B R Ambedkar National Institute of Technol	12.90	15.03	+2.13
56	Indian Institute of Technology Jammu	24.63	25.03	+0.40
57	Indian Institute of Technology, Tirupati	25.61	24.62	-0.99
58	Manipal University Jaipur	10.49	10.78	+0.29
59	Manipal Institute of Technology	10.76	12.58	+1.82
60	Madan Mohan Malaviya University of Technology	13.64	15.41	+1.77
62	Motilal Nehru National Institute of Technology	15.78	17.19	+1.41
63	Indraprastha Institute of Information Technolo	16.16	16.82	+0.66
64	Indian Institute of Technology Palakkad	20.16	21.75	+1.59
65	National Institute of Technology Delhi	22.32	20.57	-1.75
66	Sardar Vallabhbhai National Institute of Techn	15.27	17.52	+2.25
67	PSG College of Technology	9.43	10.34	+0.91
67	Sathyabama Institute of Science and Technology	9.58	10.74	+1.16
69	International Institute of Information Technol	25.89	26.01	+0.12
70	Netaji Subhas University of Technology (NSUT)	12.09	13.27	+1.18
71	Banasthali Vidyapith	11.34	14.63	+3.29
72	Indian Institute of Technology Bhilai	24.45	26.08	+1.63
73	National Institute of Technology Srinagar	16.29	17.36	+1.07
74	University of Hyderabad	14.38	22.86	+8.48
75	M. S. Ramaiah Institute of Technology	11.43	12.64	+1.21
76	Christ University	10.97	12.38	+1.41
77	Indian Institute of Technology Dharwad	19.88	20.93	+1.05
78	Rajiv Gandhi Institute of Petroleum Technology	20.26	19.26	-1.00
79	Sant Longowal Institute of Engineering & Techn	22.29	17.82	-4.47
80	Vignan's Foundation for Science, Technology an	7.21	7.62	+0.41
81	Maulana Azad National Institute of Technology	13.33	15.83	+2.50
82	National Institute of Technology, Jamshedpur	14.68	16.97	+2.29
83	National Institute of Technology Meghalaya	17.72	20.94	+3.22
84	Jain university,Bangalore	9.56	11.71	+2.15
85	National Institute of Technology Kurukshetra	13.49	15.09	+1.60
86	National Institute of Technology, Raipur	11.47	13.45	+1.98
87	Vel Tech Rangarajan Dr. Sagunthala R & D Insti	6.60	5.95	-0.65
88	AU College of Engineering (A)	19.84	19.87	+0.03
89	Chitkara University	10.67	11.38	+0.71
90	COEP Technological University	17.88	17.16	-0.72

Appendix C — CV table, selection rows

Parameter	Family	Strategy	n	CV MAE	CV RMSE	CV R ²
ESCS	Isotonic	pooled	90	0.828	1.146	0.867
ESCS	Isotonic	hybrid	90	0.853	1.168	0.862
ESCS	CappedLinear	pooled	90	0.868	1.188	0.857
ESCS	PowerLaw	pooled	90	0.868	1.188	0.857
ESCS	CappedLinear	hybrid	90	0.873	1.192	0.856
ESCS	CappedLinear	per_year	90	0.877	1.195	0.855
ESCS	PowerLaw	hybrid	90	0.879	1.183	0.858
ESCS	Isotonic	per_year	90	0.884	1.202	0.854
ESCS	PowerLaw	per_year	90	0.896	1.187	0.857
ESCS	PreviousYear	per_year	90	0.977	1.400	0.801
ESCS	LogLinear	pooled	90	1.267	1.562	0.753
ESCS	LogLinear	hybrid	90	1.273	1.568	0.751
ESCS	LogLinear	per_year	90	1.280	1.575	0.749
ESCS	MeanBaseline	pooled	90	2.748	3.174	-0.021
ESCS	MeanBaseline	hybrid	90	2.748	3.174	-0.021
ESCS	MeanBaseline	per_year	90	2.749	3.175	-0.021
FPPP	PreviousYear	per_year	92	0.275	0.464	0.948
FPPP	AdditiveIsotonic	per_year	92	0.510	0.689	0.885
FPPP	AdditiveIsotonic	hybrid	92	0.513	0.647	0.899
FPPP	AdditiveIsotonic	pooled	92	0.569	0.695	0.883
FPPP	AdditiveLog	per_year	92	0.631	0.905	0.801
FPPP	AdditiveLog	hybrid	92	0.688	0.895	0.806
FPPP	AdditiveLog	pooled	92	0.770	0.942	0.785
FPPP	MeanBaseline	per_year	92	1.543	2.055	-0.024
FPPP	MeanBaseline	hybrid	92	1.588	2.054	-0.023
FPPP	MeanBaseline	pooled	92	1.657	2.073	-0.043
FRU	AdditiveLog	per_year	89	0.987	1.447	0.924
FRU	AdditiveLog	hybrid	89	1.151	1.592	0.908
FRU	AdditiveIsotonic	per_year	89	1.156	1.520	0.916
FRU	AdditiveIsotonic	hybrid	89	1.283	1.694	0.896
FRU	AdditiveIsotonic	pooled	89	1.630	2.088	0.842
FRU	AdditiveLog	pooled	89	1.639	1.985	0.857
FRU	PreviousYear	per_year	89	3.598	3.828	0.468
FRU	MeanBaseline	hybrid	89	4.307	5.384	-0.052
FRU	MeanBaseline	per_year	89	4.313	5.317	-0.026
FRU	MeanBaseline	pooled	89	4.342	5.507	-0.100
GMS	PreviousYear	per_year	93	0.410	0.565	0.979
GMS	LogLinear	per_year	93	0.484	1.146	0.914
GMS	Isotonic	hybrid	93	0.493	1.143	0.914
GMS	PowerLaw	hybrid	93	0.509	1.228	0.901
GMS	Isotonic	per_year	93	0.510	1.167	0.911
GMS	LogLinear	hybrid	93	0.515	1.146	0.914
GMS	PowerLaw	per_year	93	0.521	1.236	0.900
GMS	Isotonic	pooled	93	0.536	1.150	0.913
GMS	PowerLaw	pooled	93	0.536	1.235	0.900

Parameter	Family	Strategy	n	CV MAE	CV RMSE	CV R ²
GMS	LogLinear	pooled	93	0.563	1.159	0.912
GMS	CappedLinear	pooled	93	0.774	1.407	0.870
GMS	CappedLinear	hybrid	93	0.782	1.401	0.871
GMS	CappedLinear	per_year	93	0.809	1.408	0.870
GMS	MeanBaseline	per_year	93	3.301	3.959	-0.027
GMS	MeanBaseline	hybrid	93	3.306	3.950	-0.022
GMS	MeanBaseline	pooled	93	3.316	3.946	-0.020
GPHD	PreviousYear	per_year	93	0.785	1.025	0.949
GPHD	Isotonic	per_year	93	1.658	2.268	0.748
GPHD	Isotonic	hybrid	93	1.660	2.253	0.751
GPHD	Isotonic	pooled	93	1.700	2.277	0.746
GPHD	LogLinear	per_year	93	1.770	2.455	0.705
GPHD	LogLinear	hybrid	93	1.795	2.465	0.702
GPHD	LogLinear	pooled	93	1.820	2.479	0.699
GPHD	PowerLaw	per_year	93	1.824	2.367	0.726
GPHD	PowerLaw	hybrid	93	1.851	2.380	0.722
GPHD	PowerLaw	pooled	93	1.881	2.396	0.719
GPHD	CappedLinear	per_year	93	1.986	2.447	0.707
GPHD	CappedLinear	hybrid	93	2.002	2.455	0.705
GPHD	CappedLinear	pooled	93	2.019	2.468	0.701
GPHD	MeanBaseline	per_year	93	3.638	4.574	-0.025
GPHD	MeanBaseline	hybrid	93	3.639	4.576	-0.026
GPHD	MeanBaseline	pooled	93	3.641	4.579	-0.028
SS	PreviousYear	per_year	93	0.680	0.954	0.930
SS	MeanBaseline	pooled	93	2.999	3.651	-0.023
SS	MeanBaseline	hybrid	93	3.000	3.652	-0.024
SS	MeanBaseline	per_year	93	3.002	3.652	-0.024
SS	AdditiveIsotonic	pooled	93	3.011	3.675	-0.037
SS	AdditiveLog	pooled	93	3.015	3.749	-0.079
SS	AdditiveLog	hybrid	93	3.031	3.752	-0.081
SS	AdditiveIsotonic	hybrid	93	3.042	3.701	-0.051
SS	AdditiveLog	per_year	93	3.048	3.758	-0.084
SS	AdditiveIsotonic	per_year	93	3.076	3.733	-0.070

Every candidate, on the common row set used for selection. cv_table_2023_2024.csv holds the wider cross-validation.